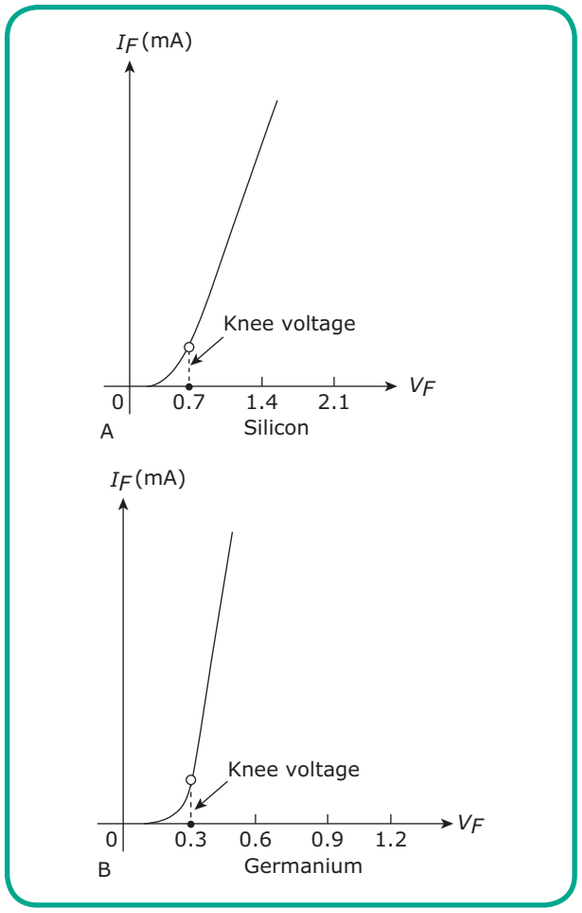


Figure 4.6 Characteristics of PN junction diode (Forward bias)

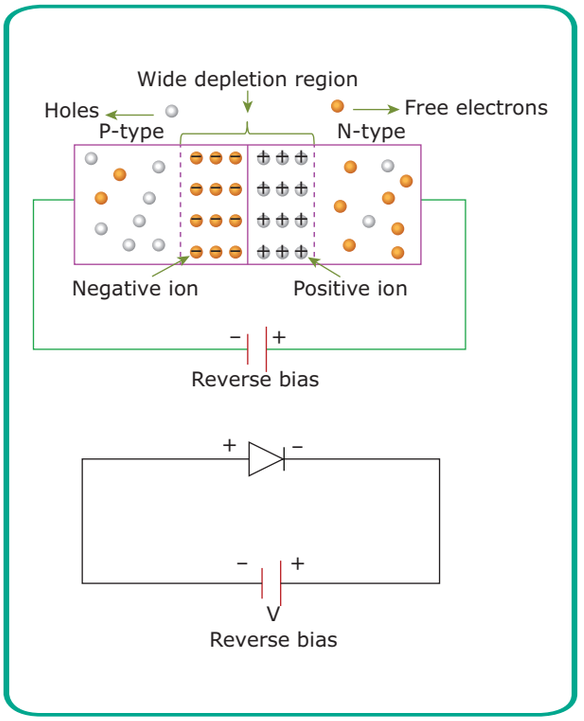


Figure 4.7 Reverse bias

4.3.5 Reverse Biasing

"When the external voltage applied to the junction is in such a direction that potential barriers is increased, it is called reverse biasing"

To apply reverse bias, connect negative terminal of the battery to p-type and positive terminal to n-type as shown in Figure 4.7. Because of increase in barrier potential the width of the depletion layer is also increased. As a result, the increased potential barrier prevents the flow of charge carriers (majority carriers) across the junction and hence the current does not flow.

From this, the students can understand the following:

1. *When pn junction is in forward biased condition, current flow occurs.*
2. *When pn junction is in reverse biased condition, no current flow occurs.*

The important term often used with *pn* junction is breakdown voltage which is explained as follows:

4.3.6 Breakdown voltage

"It is the minimum reverse voltage at which pn junction breakdown with sudden rise in reverse current".

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